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Personalized Review Selection

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Abstract

With the popularity of the online social network, reviews gradually becoming the main data source for users to understand the qualities of the goods to be purchased. However, with the proliferate of online reviews, these large amounts of reviews make it difficult for users to select useful information. Aiming to enable users to quickly obtain valid information from large amounts of reviews, this paper proposes a new method named PG to implicate personalized review selection. The proposed method in our paper is efficient for users by helping them select useful information from massive reviews.

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1. Introduction

With the sustained and rapid growth of online reviews, browsing online reviews and making decisions is a lifestyle of users [15]. By providing users with valuable sources of information, online reviews not only help them make important decisions, but also allow various online platforms to benefit and grow their business. As one

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important form of contents generating by users, online reviews describe consumers experiences including features of products which are not mentioned officially. As such, online reviews gradually become the main information source for customers [17]. [16] confirm that online product reviews help users during decision-making.

However, as online reviews increasing sharply, users have to spend a lot of time to read online reviews. Therefore, researchers pay attention to review selection and online review generation to reduce redundancy of reviews and save time for users. Users tend to be inundated with various reviews. In this case, it is unclear which reviews are valuable to users, due to two factors, the length and clutter of the reviews. If the length of one review is too long, it will be a spam review since it is difficult for users to understand [15]. Similarly, if one review contains too many different fields (with large cluster length), it can be also classified to spam review since users will hardly get accurate information [5].

[16] proposed a method to select a representative group of reviews for each product. It considered both the coverage of the properties of these products and the diversity of reviews to select useful reviews. In addition, a lot of works have demonstrated the applications of micro-reviews in the review selection. [11], [7] and [8] studied the micro-reviews and used them to conduct review selection. [7] matched the review sentences with micro-reviews, and selected a group of small reviews containing as many micro-reviews as possible. [15] developed algorithms and models for predicting the usefulness of reviews, which discovered the most helpful reviews for the given product. [6] studied and analyzed users' historical reviews and then summarized users' preferences. From the consideration of online review selection focusing on coverage, redundancy, and representativeness mostly, [4] proposed approach named RewSel, which could help consumer decision-making by extracting useful information from online reviews with high coverage and low redundancy. [13] investigated personalized review selection and updated a new algorithm named Personalized Top Reviews Sets (PTRS) based on Top Reviews Sets (TRS) algorithm, PTRS considered which products were important to users and proposed a new trade-off criteria.

However, the personalized approaches they proposed are susceptible to the following two problems. First of all, a small number of reviews may not cover all aspects of the content. Secondly, they did not consider the user preference.

In this paper, we extract aspects in each sentence by using a neural network with an attention mechanism. We optimized three goals - similarity, coverage, and personalization correlation coefficients - to select reviews that were more appropriate for different users. Our proposed method contains the following three steps:

- Extract aspects of each sentence.
- Summarize and consider each user's attention.
- Make review selection.

In the first step, the nouns in each sentence are analyzed and extracted by combining the neural network method with the attention mechanism. Then, the extracted aspect words are analyzed and clustered to represent the respective concerns of each user. Different users are represented by different vectors, and each vector is selected to represent the user's preferences. Thirdly, some vectors which are similar to the user preference vector are selected by the optimization method from all the review vectors. The main contribution of this paper is the proposed novel method can implement the personalized review selection. By comparing with traditional review selection method, the experiment result confirms a better performance.

2. Related Work

The goal of review selection is to extract a subset from all reviews of an entity (for example, a restaurant or a hotel) for different users to provide each user a set of reviews that are relevant to that user's attention. Although, as far as we know, the exact formulation of the question we raising has not been considered before. Our work is linked to existing work in the field of review summaries and review selection. Much of the work [4, 5] address the problem of selecting a subset of reviews from a given corpus. So far, all of these works have limitations. The disadvantage of review selection is that it contains redundantly and non-representative reviews. For example, a user cares about the service of a hotel, but the reviews selected in previous work may not relevant to the user concerns like 'the environment is clean, the sound is not noisy and the breakfast is good.'

Review summaries do not evaluate the immediacy and narrative structure of each review. For instance, the summaries about a mobile phone (eg, 'The phone feels good, because music has a little noisy, so you have to make a

call.’) are illogical but most younger users more care about mobile phone performance, such as the speed when playing games. In [3], the summary was a list of features that counted positive and negative opinions and some examples. [5] accurately simulated the opinion distribution in the underlying corpus, designed effective algorithms, and selected a characteristic set of reviews. The processing of reviews is helpful not only to users but also to the description of products properties. Under the requirement of providing high-quality reviews, both the diversity of opinions and the coverage of properties should be taken into consideration.

However, users are usually focused on a particular entity and multiple entities of the same neighborhood or category. [8] solved the problem with techniques by summarizing multiple entities in a collection using newly synthesized micro-reviews related to multiple collections. The representation of the set is found by identifying some aspects of common aspects connecting two or more entities in the set.

Considering the redundancy of selected reviews from the user’s point of view, [13] analyzed and selected personalized top K reviews for users according to the aspects of the product, along with what users prefer to focus on. Based on the above considerations, we focus on selecting multiple reviews which are related to the user’s concerns regarding the coverage and similarity of the reviews.

3. Methodology

3.1. Aspect Extraction

Aspect extraction is one critical step in analysis user’s preference, most of the existing works adopt the aspect extraction tends to apply variants of topic models. However, topic models usually do not produce highly relevant aspects. As such, this paper utilizes the Attention-based Aspect Extraction (ABAE, [2]) to extract aspects. ABAE can not only learn substantially higher quality aspects but also be more effective to capture the aspects of reviews than previous methods. It contains the following steps:

- Firstly, given an input, the attention mechanism is used to filter out non-aspect words by reducing the weight of non-aspect words, and then use the remaining weighted words to construct review embedding.
- Secondly, the sentence embedding reconstruction is constructed as linear combination by the aspect embedding.
- Finally, after dimension reduction and reconstruction, ABAE aims to filter reviews and maintain the key information of the reviews to the K embedded aspects.

3.2. Personalized Profile Generation

We choose the hierarchical clustering as a method of cluster analysis. Cluster analysis is one of the most common forms of unsupervised learning, which is often used to find structures in unlabeled data. We use clustering to extract unique aspects. For instance, noodles and steamed buns are both staple foods.

We use a nested hierarchical clustering tree to calculate the similarity between the different types of data points. It can stop partitioning at any time. The main steps are as follows:

- The initial state of N isolated nodes are obtained by removing all edges of the network;
- Compute the similarity of each pair of nodes in the network;
- According to the similarity from the highest to the lowest, a tree graph is formed by the connection levels of corresponding pairs of nodes;
- Based on the actual needs, the community structure is obtained by cross-cutting the tree.
- By calculating the Dunn Index (DI) (a metric for assessing clustering algorithms, an internal evaluation scheme.) in clustering, the categories of clustering are obtained. The number of categories will be further applied to represent the personalized profiles of each user by vectors.

After above clustering steps, we get the categories for all aspects. In order to get a user preference vector, we count all the aspects of a user’s historical reviews and normalize them.

3.3. Review Selection

3.3.1. Problem Definition

We define a vector $\mathbf{u}$ as the user’s personal profile, and a vector $\mathbf{r}$ as one review’s embedding. $\mathbf{r} \in \mathbf{R}$, and $\mathbf{R}$ is the set of all reviews. The problem of personalized review selection is then transformed into the task of selecting a set of reviews that users care about from $\mathbf{R}$, and these reviews should be with large similarity and comprehensive coverage. It can be represented as a review vector, in which each element have two labels, one represents that the

corresponding review is selected and another represents that it is not selected. Given a vector $X = [x_1, x_2, \dots, x_n]$, The selecting function F is defined as follows:

$$F(r, x) = \begin{cases} 1 & \text{if } r \text{ is selected,} \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

A review vector $r \in R$ is selected by user preference vector $u \in U$. Given a set of reviews R and users' reviews U , and the selecting function F . We define the set of reviews that are select to the user as R_u :

$$R_u = \{r \in R, u \in U, \exists x \in X, F(r, x) = 1\} \quad (2)$$

Note that we define $len(R_u)$ as the length of the selected reviews.

3.3.2. Coverage

Coverage is a value that must be calculated in a review selection problem, and in our work, it is also an optimal goal for making personalized review selection for users. Coverage is calculated by the review vector r and user preference vector u . Firstly, the XOR operation is carried out between the user preference vector u and the review vector r . We choose $XOR = 0$, the user preference vector and the review vector are from the same aspect category. We define the coverage value z as the result of XOR operation.

$$z(r, u) = |u \oplus r| \quad (3)$$

Next, we define coverage $Cov(r, u)$ of a review is represented by the fraction of z to the total review vector dimension $dim(r)$. Based on above definitions, the average of all selected $Cov(r, u)$ can be defined as:

$$Cov(r, u) = \frac{z(r, u)}{dim(r)} \quad (4)$$

$$Cov(R, u) = \frac{\sum_{r \in R} Cov(r, u)}{len(R_u)}, r \in R \quad (5)$$

3.3.3. Similarity

A review vector and a user preference vector are similar if they share the same important aspects. The *Earth Movers distance (EMD)* [[9], [10]] is used to measure the difference between two probability distributions over a domain. [14] make use of the *EMD* as the scoring function to measure the diversity between the two groups. In order to evaluate the similarity between the review vector and the user preference vector, an appropriate function $Sim(\bullet)$ that can measure the difference between two vectors is required. Therefore, *EMD* is adopted to evaluate the similarity of the users' preferences and reviews. The degree of Similarity between r and u can be measured as follows:

$$Sim(r, u) = EMD(r, u) \quad (6)$$

3.3.4. Personalization of User (P_u)

We define *Personalization of User* (P_u) to measure if the users are in given reviews. Since aspect distribution shows users' personalization, it is necessary to define a personalized optimal goal based on user personalization. The higher relevance of the selected reviews to user's personalization, the better of the method. Therefore, we use the Spearman as the correlation between the user preference vector and the restaurant review vector. Finally, we introduce P_u to calculate the average relevance of selected reviews.

$$P_u(R, U) = \frac{\sum_{r \in R} Cor(r, u)}{len(R_u)}, r \in R, u \in U \quad (7)$$

A high P_u indicates that a user will obtain richer information about the review aspects that are important to this user.

3.3.5. Personalized Review Selection

Based on the above concepts, we define personalized review selection as a multi-objective optimization task. we want to select the reviews that have a higher *coverage*, *similarity*, and the larger P_u . We also need to limit the number of reviews selected for users, so we propose a multi-objective optimization problem. $x_i \in X, (i = 1, 2, 3, \dots)$, x_i indicates whether the review can be selected, and values 0 or 1. And the problem as a set of constraints:

$$\max \{F_1, F_2, F_3\} \quad (8)$$

$$\begin{cases} F_1 = \text{Sim}(r, u) \\ F_2 = \text{Cov}(R, u) \\ F_3 = P_u(R, U) \end{cases} \quad (9)$$

$$\text{s.t. } k = \sum_{i=1}^n x_i \leq \varepsilon, i = 1, 2, 3, \dots, n \quad (10)$$

We define a *Personalized Genetic (PG)* algorithm based on genetic algorithm considering above optimizations. The best set of reviews are produced by maximizing the three targets, namely, *Coverage*, *Similarity*, and P_u . The details of the algorithm are shown in Algorithm 1. *PG* algorithm requires: the solution domain is obtained by genetic iteration, then we use the fitness function represented by the optimization to select the result.

Algorithm 1 PG Algorithm

Input:

Set of reviews R and U ;

Output:

A set of reviews R of size K .

- 1: Generate the initial population R
 - 2: Set the number of iterations
 - 3: Calculate the value of $F_1 \leftarrow \text{Sim}(r, u)$
 - 4: Calculate the value of $F_2 \leftarrow \text{Cov}(R, u)$
 - 5: Calculate the value of $F_3 \leftarrow P_u(R, U)$
 - 6: Fitness $F = \max\{F_1, F_2, F_3\}$
 - 7: Fitness evaluation of the initial population
 - 8: **while** not meeting the number of iterations **do**
 - 9: Select a pair of parents
 - 10: Crossover
 - 11: Mutation
 - 12: **end while**
 - 13: **return** Retain the best set of reviews R
-

We have all the review vectors to a genetic performance of the solution domain, and use the binary code. Therefore, we translate the review selection problem as a 0–1 knapsack problem [1] of objective optimization in genetic algorithm. We introduce three objectives, *Coverage*, *Similarity*, and P_u , as a fitness function to evaluate the solution domain. Once we define the input and output of *PG*, the *PG* continues to generate the initial population and the to optimize it by the application of the selection, crossover, and mutation operators.

4. Experiment and Analysis

4.1. Data Collection

Reviews used for this study were collected from Yelp.com. We choose the data domain about restaurants since it is a popular domain with active review platforms. We collect the top 160 Hong Kong restaurants with the highest number of reviews in November 2018 from Yelp.com. It includes 2659 users and 89350 reviews. There are three features in experimental data, restaurant name, user name, and review content. These restaurants are all registered in Yelp.com on Hong Kong. Each user has actually signed up from the site, and the reviews are real-time impressions of each restaurant.

The reviews we choose to give user are some simple sentences rather than whole paragraphs. So, we split these long reviews to short sentence first, because the users are more like to read short reviews. After that, we extract aspect from each review to make review selection.

4.2. User's Preference

For selection the reviews that can meet the users personalization, we collect the profile of each user and remove duplicate aspects of each review first. Then, we take the hierarchical clustering to analyze the aspect of each user. Fig. 1a shows that the results of hierarchical clustering. And the number of clusters maximized by *DI* is considered to be the optimal number of clusters *k*. In Fig. 1b, we can see that the *DI* should be the maximum value (distance is 55), but in order to more accurately represent the user's personalization, we choose the distance is 50.

From Fig. 1c, we can get 7 categorical aspects. Aspects of each review in a restaurant carry out statistical and normalized processing of the clustering results, which is expressed as the restaurant review vectors. The same statistical and normalized processing is carried out for all aspects in a user's historical reviews. A seven-dimensional vector represents a review, and the degree of each dimension is the particular aspect which a user is interested in. The numbers in the different dimensions of the vector are the user's attention to that category. For example, A user's attention in the first category is 0.33, the third category is 0.5, and the seventh category is 0.27, the other is 0. Then the vector can be [0.33,0,0.5,0,0,0,0.27].

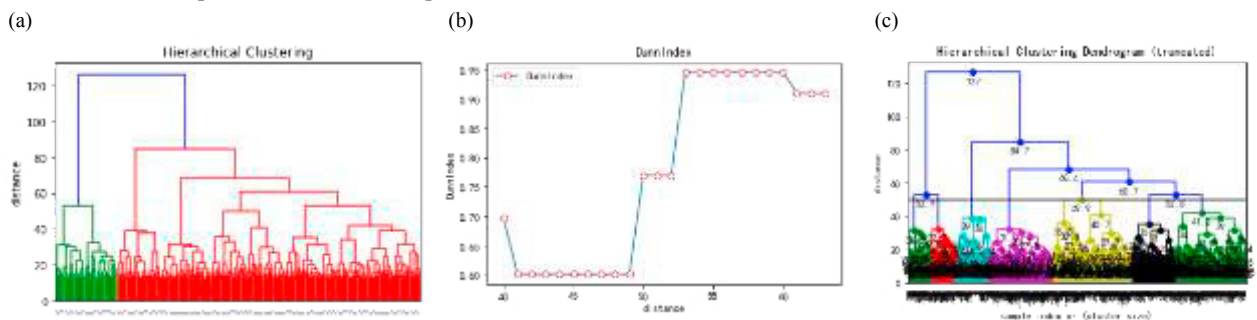


Fig. 1. (a) Preliminary clustering results; (b) Dunn Index(DI); (c) The results of hierarchical clustering

4.3. Personalized Review Selection

After the restaurant review vectors and the user preference vectors are prepared, the next step is to use the genetic algorithm to select the personalized review for the user. The three optimize objectives in our experiments are, *Coverage*, *Similarity*, and P_{re} . Our comparison is from the change in the numbers of iterations to observe the *Coverage* of the three objectives. We compared the impacts of *Similarity*, P_{re} , as well as *PG* algorithm on *Coverage*.

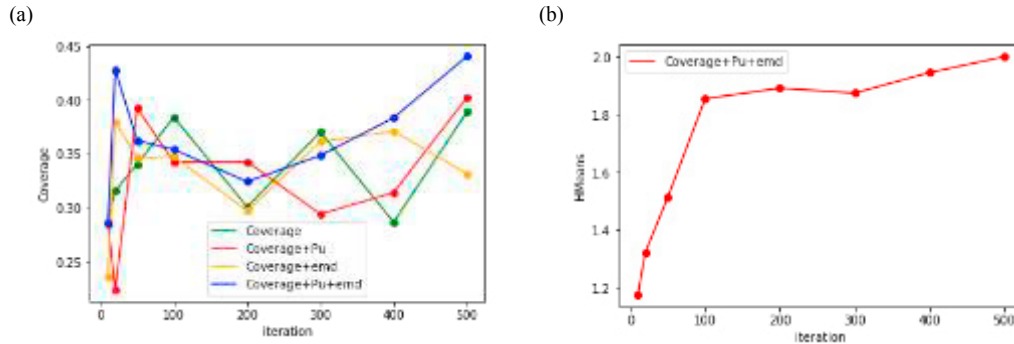


Fig. 2. (a) Comparing coverage and others; (b) HMeans.

From the Fig. 2a we can find that when the number of iterations is small, the results calculated by the algorithm are all local optimal values. Generally, when the number of iterations increases, the coverage of PG algorithm is the largest. With the increase of coverage and similarity, the reviews are closer to users' concern. The larger the P_u , the richer review information we can select for each user.

Personalization of User (P_u) in information retrieval, we use the harmonic mean (In mathematics, the harmonic mean is one of several kinds of average.):

$$HMeans(R) = \frac{2 * P_u(R, U) * Sim(r, u)}{P_u(R, U) + Sim(r, u)}, r \in R, u \in U \quad (11)$$

To emphasize the effect of PG at achieving both P_u and *Similarity*, Fig. 2b plots the harmonic mean. It shows that with the increase in the number of iterations, the efficiency of the algorithm is more pronounced.

5. Conclusions

Due to different users with different needs, too many redundant spam reviews will waste time for users to read. Therefore, review selection for users is essential. In order to find a group of personalized reviews for each user indifferent entity environments, we proposed the concept of personalized review selection, which is a novel efficient method considering coverage and personalized of target reviews. Our work introduces the personalization of users through aspects in user reviews, and proposes a PG algorithm, which can select different reviews to different users, and consider more personalized aspects users care about. It includes not only reviews of high similarity and coverage, but also more focused on review aspects that are important to the users. The experiments of a corpus of restaurant reviews and user preferences show that our approach is effective and useful.

Furthermore, the implementation of personalized review selection makes the product improve in a better direction. Users can save a lot of time when browsing web sites, and the interests of merchants can be maximized to make changes according to the needs of different reviews. Besides, the personalized review selection method we proposed in this paper can also be used to handle other commodity reviews, as well as hotels. In the future, we plan to use the multi-objective optimization algorithm (*Particle Swarm Optimization* [12]) for improving the results of review selection.

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